

Model Performance Cheat Sheet

Classification & Regression Metrics

Notation

y_i : true label/value, $\hat{y}_i$: predicted label/value, $\hat{p}_i$: predicted probability

n : number of samples, p : number of features

TP/TN/FP/FN: True/False Positive/Negative counts

Probabilistic & Ranking Metrics

$$\text{ROC AUC} = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR} \quad (8)$$

$$\text{PR AUC} = \int_0^1 \text{Precision}(\text{Recall}) d\text{Recall} \quad (9)$$

$$\text{Log Loss} = -\frac{1}{n} \sum_{i=1}^n [y_i \ln(\hat{p}_i) + (1 - y_i) \ln(1 - \hat{p}_i)] \quad (10)$$

$$\text{Brier Score} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2 \quad (11)$$

Classification Metrics

Accuracy & Error

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Balanced Acc} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (2)$$

Precision, Recall & Specificity

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (5)$$

F-Scores

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

$$F_\beta = \frac{(1 + \beta^2) \cdot \text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}} \quad (7)$$

Multi-class Averages

Macro: Mean of per-class metrics

Micro: Pool all counts, then compute metric

Weighted: Class-frequency weighted macro average

Chance-Corrected Metrics

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (12)$$

When to Use Classification Metrics

Imbalanced data: F1, PR AUC, MCC

Probability calibration: Log Loss, Brier Score

Threshold tuning: ROC/PR curves

Cost-sensitive: Custom metrics with cost matrix

Regression Metrics

Error Magnitude

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (13)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (14)$$

$$\text{RMSE} = \sqrt{\text{MSE}} \quad (15)$$

$$\text{MedAE} = \text{median}(|y_i - \hat{y}_i|) \quad (16)$$

Relative & Percentage Errors

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i + \epsilon} \right| \quad (17)$$

$$\text{sMAPE} = \frac{100}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{\frac{|y_i| + |\hat{y}_i|}{2} + \epsilon} \quad (18)$$

Goodness-of-Fit

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (19)$$

$$\text{Adj } R^2 = 1 - (1 - R^2) \frac{n-1}{n-p-1} \quad (20)$$

$$\text{EVS} = 1 - \frac{\text{Var}(y - \hat{y})}{\text{Var}(y)} \quad (21)$$

Scale-Free & Cross-Series

$$\text{nRMSE} = \frac{\text{RMSE}}{\text{range}(y)} \text{ or } \frac{\text{RMSE}}{\bar{y}} \quad (22)$$

$$\text{MASE} = \frac{\text{MAE}}{\frac{1}{n-1} \sum_{t=2}^n |y_t - y_{t-1}|} \quad (23)$$

Robust & Quantile Loss

Huber Loss_δ: Quadratic if $|r| \leq \delta$, linear otherwise

Pinball Loss_τ: τr if $r \geq 0$, $(\tau - 1)r$ if $r < 0$

When to Use Regression Metrics

Outliers present: MAE, MedAE, Huber Loss

Cross-dataset comparison: nRMSE, MASE

Forecasting: sMAPE, MASE

Quantile prediction: Pinball Loss

Quick Reference Tables

Binary Classification

Metric	Formula
Accuracy	$(TP + TN)/(TP + TN + FP + FN)$
Precision	$TP/(TP + FP)$
Recall	$TP/(TP + FN)$
Specificity	$TN/(TN + FP)$
F1-Score	$2 \cdot \frac{\text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}}$
MCC	See formula above
Log Loss	See formula above
Brier Score	$\frac{1}{n} \sum (\hat{p}_i - y_i)^2$

Regression

Metric	Formula
MAE	$\frac{1}{n} \sum y_i - \hat{y}_i $
RMSE	$\sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$
MAPE	$\frac{100}{n} \sum \left \frac{y_i - \hat{y}_i}{y_i + \epsilon} \right $
R^2	See formula above
MASE	See formula above